

On the ubiquity of temperature response asymmetry

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Abstract

The biological response to temperature is almost universally asymmetric: rates change slowly on the cold side of the optimum and steeply on the warm side. We argue this temperature response asymmetry (TRA) recurs across scales, from enzyme kinetics through ectotherm metabolism, human mortality, peak electricity demand, and macroeconomic productivity. Three recent findings now make a unified cross-scale treatment tractable. A meta-analysis of 2,700 thermal performance datasets across seven kingdoms shows collapse onto a single rescaled curve. Stochastic thermodynamics demonstrates that heating is intrinsically faster than reciprocal cooling far from equilibrium. A transition-state generalisation of Arrhenius kinetics derives the asymmetry from conformational entropy and extends the same formalism to socioeconomic aggregates. A recent PNAS exchange is actively contesting what "universality" means at the biological scale. Our Perspective engages with that exchange while extending its underlying question to socioeconomic systems that have no direct enzymatic substrate, asking whether the asymmetry has a single physical origin propagating upward through scales or arises convergently from distinct mechanisms. The stakes are direct: projected 21st-century mortality shifts will shorten average human lifespan by months, and historical greenhouse-gas emissions already commit the global economy to a 19% income reduction by mid-century, both driven by the steeper warm-slope side of the asymmetric response.

Keywords thermal performance curves, stochastic thermodynamics, climate impacts

The first information that we usually get from weather forecasting is the temperature. This is because temperature is such an important environmental parameter that determines our physiological comfort. In fact, temperature is one of the main factors that influence many processes in physics, chemistry, and biology. Kelvin, the unit used to measure thermodynamic temperature, is one of the seven base units of the *Système international d'unités*.

Our everyday experience tells us that plants die if they get too cold or are exposed to too much heat; between the two extremes, there is a temperature at which they thrive. Intuitively, one would expect such an optimum to exist. However, it is less intuitive that the temperature response differs for heat and cold, as we discuss below. More generally, the rate of change of the response is slower on the cold side and faster on the warm side, regardless of whether the curve is concave or convex. Here we argue that an extraordinary spectrum of complex systems, from enzyme kinetics and human physiology to global socioeconomic behavior, exhibits the same characteristic temperature response asymmetry (TRA).

Let us begin with the effect of temperature on human physiology. Many epidemiological studies estimate temperature mortality curves that provide the relative risk as a function of ambient temperature. The association shows that the risk increases slowly for cold temperatures and generally escalates quickly at high temperatures (e.g. Gasparrini *et al.*, 2015). Although this signature is well-documented for many cities in different countries and climates, there is no universal temperature of minimum mortality Krummenauer *et al.* (2019). However, due to the steeper heat-slope, projections for the late 21st century show that the rise in heat-related deaths is expected to fully override the benefits of reduced cold-related mortality Ballester *et al.* (2011); Masselot *et al.* (2023).

The relationship between ambient temperature and human affective states is characterized by a non-linear distribution, where positive sentiment peaks at moderate thresholds Baylis (2020); Moore *et al.* (2019). Although physiological signals of discomfort serve as a protective warning system against thermal extremes, subsequent sentimental adaptation seems asymmetric. Evidence suggests that while conscious tolerance of high-temperature environments may increase over time, the sub-perceptual negative impact on well-being does not diminish. Consequently, chronic exposure to heat induces a persistent, unmitigated emotional tax that bypasses standard habituation mechanisms.

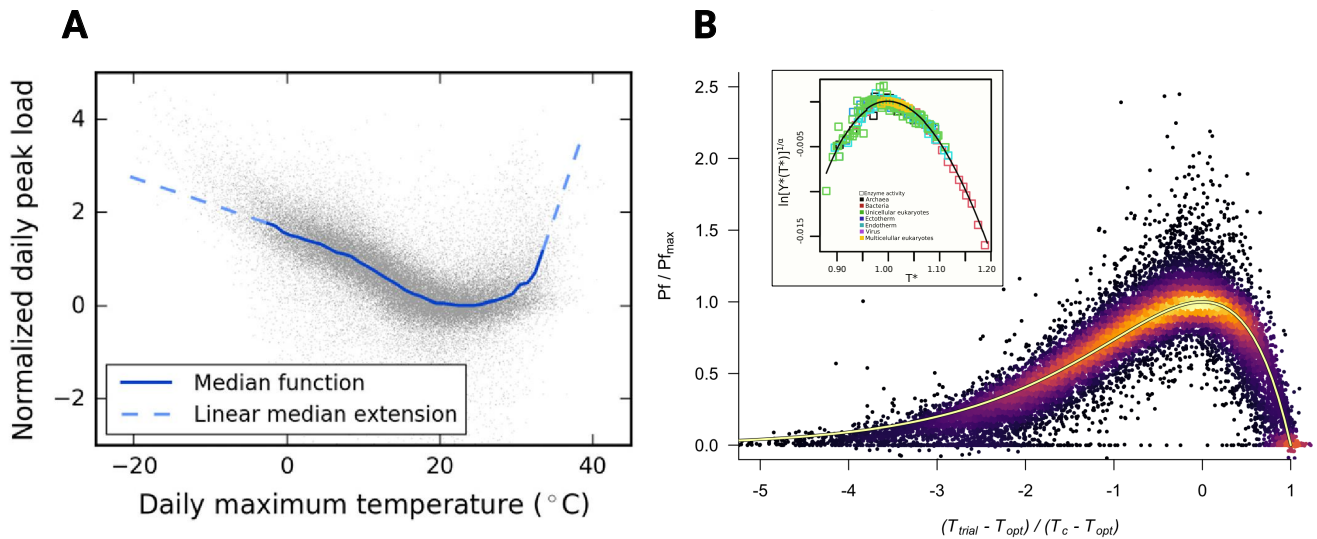


Figure 1 Temperature response asymmetry across scales. **A.** Electricity consumption (peak load) in the US, showing the characteristic asymmetry in socioeconomic energy demand as a higher-level consequence of biological constraints Wenz *et al.* (2017). **B.** Universal collapse of $\sim 2,700$ thermal performance datasets rescaled by optimal and critical temperatures Arnoldi *et al.* (2025). The inset shows enzymatic activity described by transition state theory, highlighting the entropic drivers of the asymmetric curve Arroyo *et al.* (2022). In both panels the warm side of the response is steeper than the cold side, reflecting the same underlying thermodynamic asymmetry regardless of whether the quantity peaks or troughs at the optimum.

Other effects of temperature related stress on human behavior are more indirect. Importantly, this biological asymmetry propagates to higher levels of organization, impacting human-engineered infrastructure as a macro-scale consequence (Fig. 1A). An example is daily electricity demand, particularly peak load, which shows a consistent asymmetry in the correlations with temperature. Empirical evidence strongly suggests that the high-temperature end of the curve is significantly steeper Klein *et al.* (2011); Wenz *et al.* (2017). The hot end of the curve is almost entirely driven by the rapid collective deployment of air conditioning (A/C) Wenz *et al.* (2017); Vicedo-Cabrera *et al.* (2021). It should be noted that the waste heat from the A/C has an amplifying effect Colelli *et al.* (2023). Moreover, a blackout during a heatwave, by turning off the A/C, would sharply exacerbate the original health risk Colelli *et al.* (2023). The significance of air conditioning extends well beyond acute crises. The modern settlement of the “sun belt” of the United States was in large part made possible by the technological capacity to cool interior spaces Hill *et al.* (2013, 2022). In regions where ambient temperatures regularly exceed body temperature, energy and monetary expenditures on cooling can equal or exceed those on heating in cold climates, itself a striking signature of TRA translated into economic terms. With ongoing warming, environmental temperatures in parts of Arizona, the Arabian Peninsula, and South Asia are already approaching thresholds that are lethal to humans without substantial energy expenditures to mitigate them. Another example is human migration. The implicit temperature response curve for migration is not gradual, but represents a critical threshold response. This threshold seems to vary based on local adaptive capacity, household vulnerability, and the political context Schlenker and Roberts (2009). Migration often occurs only after a climatic event crosses a critical point of socio-economic vulnerability, such as the absolute loss of income Abel *et al.* (2019).

This leads to the question, to what extent does temperature affect economic productivity? The seminal work by Burke *et al.* (2015) overturned previous linear assumptions by identifying a productivity function that exhibits an optimum at an annual average temperature of approximately 13°C . Below this optimum, warming may marginally benefit productivity; above it, productivity declines at an accelerating rate. Economic productivity includes agricultural output, clearly demonstrating how crossing a physiological threshold translates into a sharp, non-linear economic damage function Schlenker and Roberts (2009); Hogan and Schlenker (2024). Crop yields for staples such as corn and soybeans show an asymmetric temperature response, with a catastrophic decrease once critical heat thresholds (typically 27°C to 30°C) are reached Schlenker and Roberts (2009). This TRA mirrors the thermal performance curves of enzymes and crops, suggesting that macroeconomic systems are bounded by the same thermodynamic constraints as their biological components Burke *et al.* (2015).

The mathematical generality of these patterns is demonstrated by the massive meta-analysis of $\sim 2,700$ empirical thermal performance measurements conducted by Arnoldi *et al.* (2025), spanning seven kingdoms and diverse metrics, from individual physiology to population growth rates (Fig. 1B). Through an algebraic rescaling of two established temperature-dependent models, this work reveals a ubiquitous exponential scaling that creates a mathematical attractor toward a universal thermal performance curve, approximated by the elementary equation $y(x) \approx e^x(1 - x)$. This remarkable collapse implies that seemingly

disparate TRA across the tree of life approximate rescaled versions of the same curve Barenblatt (1996). A critical consequence is the trade-off between peak performance and thermal tolerance, whereby warmer-adapted systems with higher optimal temperatures tend to have narrower thermal breadths, making them more sensitive to temperature fluctuations Arnoldi *et al.* (2025). The interpretation of this collapse – whether universality reflects a specific underlying mechanism or a generic asymptotic property – has recently become the subject of active debate within biology Pawar and Kontopoulos (2025); Arroyo *et al.* (2026); Arnoldi *et al.* (2026).

Temperature-dependent quantities display a characteristic response that can appear either as a concave curve, with a maximum at the optimum (e.g. enzyme reaction rates, metabolic rates of ectotherms, population growth rates, species richness, economic productivity), or as a convex curve, with a minimum at the optimum (e.g. metabolic rates of endotherms, mortality rates, energy consumption in cities, development time, generation time, and biological timescales in general). These two shapes are not fundamentally different phenomena. They are dual mathematical expressions of the same underlying thermodynamic process, related by a simple inversion Arroyo *et al.* (2022, 2025). Whether the curve opens upward or downward depends on which quantity is being measured – a performance rate or its reciprocal cost – rather than on a distinct mechanism.

Two distinct questions therefore arise. First, why is the temperature response so general, manifesting itself from the molecular to the socioeconomic scale? And second, why is it commonly asymmetric, and what is the origin of that asymmetry? By asymmetry we mean that the absolute temperature response is smaller below the optimum temperature than above, $\left| \frac{dX}{dT} \right|_{T \ll T^*} < \left| \frac{dX}{dT} \right|_{T^* \ll T}$, where X is the considered measure and T^* is the respective optimum or pessimum temperature. Here, we want to discuss two possible explanations of the asymmetry, a recent finding from physics and an established debate in biology. We begin with the former. First, recent advances in stochastic thermodynamics offer a foundational explanation for kinetic asymmetry in systems pushed far from equilibrium Ibáñez *et al.* (2024). This empirical Thermal Kinematics framework, established on optically trapped colloidal particles undergoing rapid temperature quenches, reveals that heating is intrinsically faster than the reciprocal cooling for any fixed pair of temperatures, challenging the assumption that systems trace the same path during reciprocal processes. The asymmetry persists even when comparing thermodynamically equidistant temperatures. A colder system, despite being statistically farther from equilibrium, heats up faster than a corresponding hotter system cools down Ibáñez *et al.* (2024). This physical finding of faster heating represents a constraint suggesting that complex systems are inherently less capable of rapidly dissipating energy/entropy gains than they are of accommodating losses.

Second, while the Arrhenius equation is a staple for modeling biological rates, it fails to account for the sharp decline in performance observed at high temperatures (Fig. 1B inset). Addressing this limitation, Arroyo *et al.* (2022) derived a first-principles theory based on Eyring-Evans-Polanyi transition state theory. The framework identifies predictable temperature-induced changes in conformational entropy (ΔS) as the primary mechanistic driver of biological rate variation. By incorporating this entropic factor, the model yields a modified exponential function that successfully quantifies the rapid kinetic acceleration toward a thermal optimum and the subsequent sharp decline observed at supra-optimal temperatures. More importantly, it provides a unified master expression applicable across biological scales, from molecular kinetics to macroscopic metabolic rates. Notably, the model is not restricted to strictly biological systems. The equation derived by Arroyo *et al.* shows that the decay of temperature-dependent quantities is captured by a scaling function of the kind commonly encountered in critical phenomena, suggesting deep structural parallels between thermal performance collapse and phase-transition physics Arroyo *et al.* (2025). Additionally, this steep asymmetric crash defines a biological hard limit that may be thermodynamically interpreted as the rapid increase in the number of non-functional configurational micro-states of the protein leading to denaturation and the critical under-performance of binding affinity as heat increases beyond the optimum Arroyo *et al.* (2022).

An independent, physiological perspective complements these thermodynamic and kinetic explanations. From the standpoint of organismal energy budgets, all assimilated energy (A) must be partitioned between respiration (R), which fuels maintenance and biological work, and production (P), which builds the biomass of surviving offspring Brown *et al.* (2004). Both R and P exhibit temperature dependence, yet they need not follow the same trajectory. For an organism to persist, lifetime production must exceed lifetime dissipation. Empirical evidence suggests that organisms have evolved so that $R > P$ at thermal extremes, while $P > R$ over an intermediate range that is shifted asymmetrically toward higher temperatures. In qualitative terms, this is the very asymmetry whose generality the present article attempts to demonstrate. An important and underexplored empirical question is how narrowly constrained the temperature of maximum net production ($P - R$) actually is. Relevant data exist for the optimal growth of cultivated plants, domestic animals, cultured microbes, and the productivity of coral reefs and tropical forests Brown *et al.* (2018), yet a systematic cross-taxon comparison has not been attempted.

The clustering of body temperature in endothermic mammals within the narrow window of approximately 37–42°C offers a telling mechanistic illustration. Classic Schowalter-Irving models of energy balance and temperature regulation show that when body temperature exceeds ambient temperature ($T_b > T_a$), thermoregulation proceeds through passive avenues of heat loss – conduction, convection, radiation – that can be modulated by adjusting the insulation of fur or feathers, with very little additional metabolic cost. When, however, ambient temperature exceeds body temperature ($T_a > T_b$), the only effective means of dissipating excess heat is evaporative water loss through sweating or panting, which entails a substantial increase in respiration rate and thereby generates yet more heat that must be dissipated. This built-in directionality of thermoregulatory cost is itself an expression of TRA at the organismal scale.

A key conceptual challenge remains in understanding how these micro-asymmetries scale up to macro-scale responses. Biological systems, such as the human body, actively regulate their internal temperature through complex feedback loops. While the underlying thermodynamic principles observed at the microscale Ibáñez *et al.* (2024); Arroyo *et al.* (2022) seem responsible for the temperature-related transitions, active regulation adds a layer of biological complexity. It is plausible that TRA persists because feedback

mechanisms are easier to maintain in a cooling regime – where the system simply facilitates dissipation – compared to a heating regime, where regulation must actively counteract runaway entropic decay and protein denaturation once the optimal threshold is breached. Mathematically, the superposition of small-scale processes can be expressed as convolution, see e.g. Burke *et al.* (2015). Via integration, the macroscale effect comprises all the contributions from micro-scale effects, which are stochastically shifted. This way, e.g. mortality could be expressed in terms of individual critical values that follow a specific distribution Prahel *et al.* (2016). Explicitly acknowledging this scale transition is fundamental to reconciling physical kinetic theory with physiology.

Thoughts on bridging the scales, however, deserve caution. The argument that micro-scale thermodynamic asymmetries *cause* the macro-scale patterns rests, at this stage, on plausibility and phenomenological analogy rather than on a rigorous derivation that traces the causal chain across each level of organization. Convergent response shapes may arise from shared physical constraints, but they may also emerge from distinct mechanisms at each scale, a possibility that only systematic cross-scale experimentation can resolve. The universality of the rescaled collapse Arnoldi *et al.* (2025) is itself phenomenological. It demonstrates a common functional form without, by itself, implying a common causal origin. This question is currently the focus of an active exchange in PNAS. Arroyo *et al.* (2026) argue that their first-principles transition-state derivation Arroyo *et al.* (2022) already accounts for the observed universality. Arnoldi *et al.* (2026) respond that universality concerns leading-order asymptotic behavior largely independent of mechanistic detail. Pawar and Kontopoulos (2025) contest the empirical universality itself at the level of parametric model fitting. The exchange turns on whether the asymmetric collapse within biology demands a specific thermodynamic origin or arises generically. Our Perspective sits adjacent to this debate rather than within it, asking whether the same characteristic shape propagates outward to scales the biological mechanisms cannot reach. Distinguishing true universality from superficial resemblance will require explicit tests, for instance whether the same scaling exponents and collapse quality hold for socioeconomic variables that have no direct enzymatic substrate. Furthermore, physiological homeostasis introduces regulatory buffers – behavioral thermoregulation, circulatory redistribution, acclimation – that can modulate, delay, or even locally invert the raw thermodynamic asymmetry. The extent to which these buffers preserve or reshape the underlying asymmetry remains an open empirical question, particularly for endotherms whose internal temperature is actively defended. An open theoretical question is whether the modification of Arrhenius-type kinetic theory by Arroyo *et al.* (2022, 2025) is sufficient, on its own, to explain both the generalized asymmetry *and* the specific shape of the response, including the value of the temperature at which the difference between anabolic and catabolic rates is maximized. The empirical generality of TRA appears convincing, but a complete theoretical account must eventually integrate transition-state thermodynamics with the energetic constraints of organismal fitness.

As the concept of temperature matured over centuries, evolving from a qualitative sense of 'mixture' to a rigorous thermodynamic ratio scale defined by linear laws McCaskey (2020), the mitigation-adaptation tradeoffs demonstrate, this linear thermodynamic definition is insufficient for characterizing the impact of thermal variance on complex systems. While the input (ambient temperature) may scale linearly, the systemic output (electricity demand, physiological stress) exhibits a profound non-linearity and asymmetry. To understand biological and socio-economic networks, we must move beyond the assumption of a neutral, linear input and account for the characteristic TRA. Such a framework must acknowledge that in systems pushed far from equilibrium – whether enzymatic or economic – heating is intrinsically more disruptive than cooling, necessitating a metric that quantifies risk as an inherent property of the thermal gradient.

The comparison of the various cases where temperature response exhibits similar asymmetry leads to the question to what extent are all manifestations of the same, possibly universal, phenomenon? An immediate follow-up question is whether its origin is physical or biological? Since most of the research is conducted in disciplinary silos, these fundamental questions do not receive the attention they should. Moreover, we see two scientific worlds that more or less independently research TRA, namely macroscopic effects on how humans react to temperature, including mortality risk, sentiment, electricity demand, and economic productivity, and microscopic effects stemming from physics and micro-biology Arroyo *et al.* (2025). Here we argue only with plausibility that microscopic processes cause the macroscopic effects. Consequently, more complex systems research – across systems, scales, and disciplines – and a rigorous framework are necessary to explain how one translates into the other.

In any case, we see great relevance, given that e.g. the structural change in mortality seasonality is projected to result in a net drop in average human lifespan of up to 3 to 4 months by 2070–2100 Ballester *et al.* (2011). The mechanisms behind the TRA are not just academic – they are of high societal relevance. E.g. high-resolution analyses have reinforced the severity of TRA, demonstrating that due to historical greenhouse-gas emissions, the global economy is already committed to a 19 % income reduction by 2050, with damages driven largely by the asymmetric impact of temperature variability and precipitation extremes Kotz *et al.* (2024).

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